15) $\nabla f\big|_{P_0} = 2xi + 2j + 2k(1,1,1/2) = 2i + 2j + 2k$ $\qquad \nabla y = j$

$$v = \begin{vmatrix} i & j & k \\ 2 & 2 & 2 \\ 0 & 1 & 0 \end{vmatrix} = 2i + 2k \Rightarrow \text{linha } t_y$$

$x = 1 - 2t,\ y = 1,\ z = \frac{1}{2} + 2t$